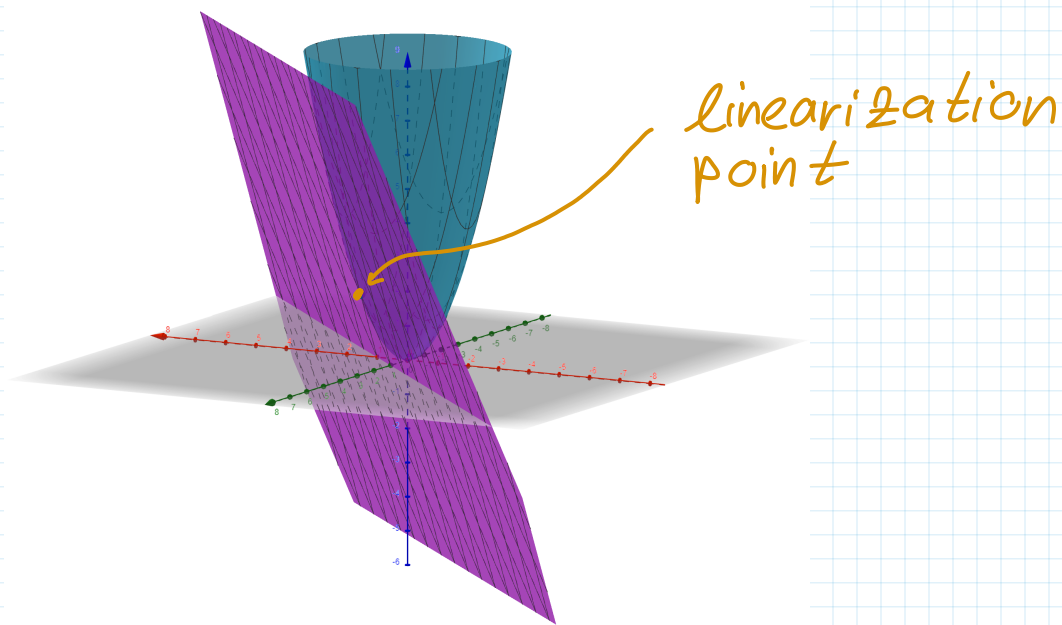


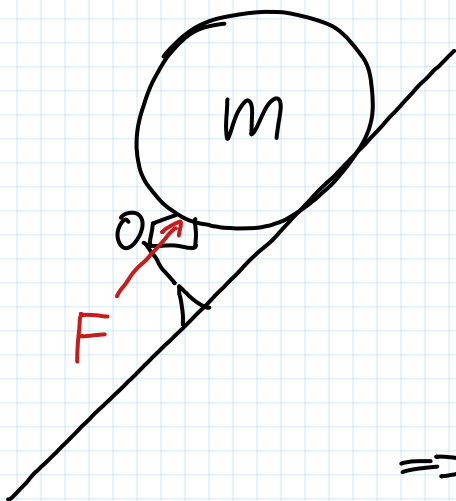
Where we left off...

$\Rightarrow$ Real world systems are almost always nonlinear.

$\Rightarrow$ In a certain "operational neighborhood" they can be approximated by linearization



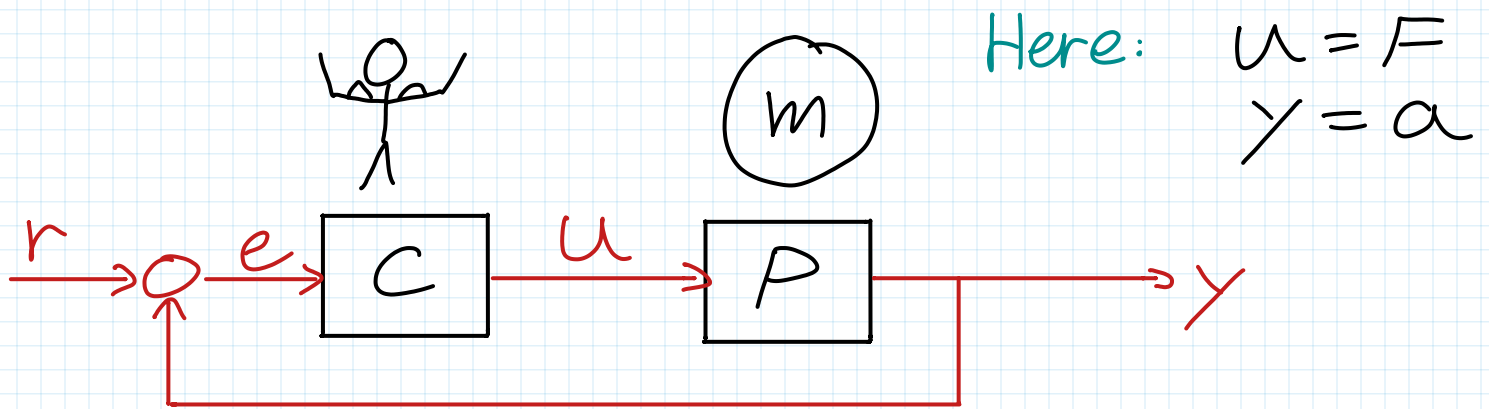
• But some systems contain nonlinearities that are hard to neglect...



In theory, Sysiphus can always stabilize the boulder but in practice he can enact only limited forces!

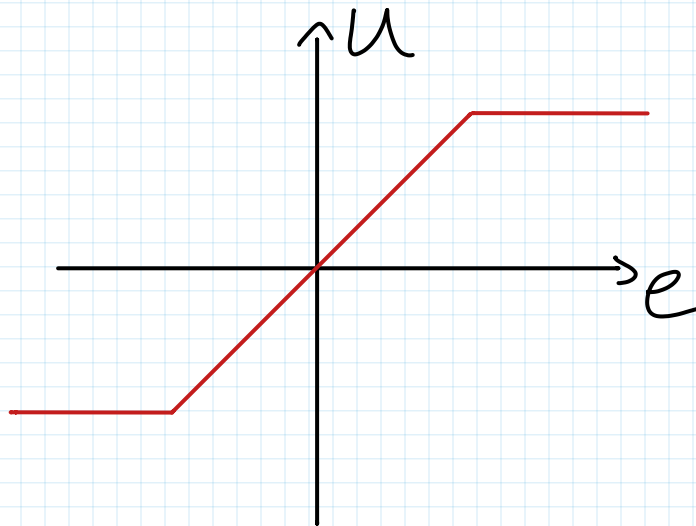
$$\Rightarrow F \in [0, 100N]$$

• This is called a saturation nonlinearity.

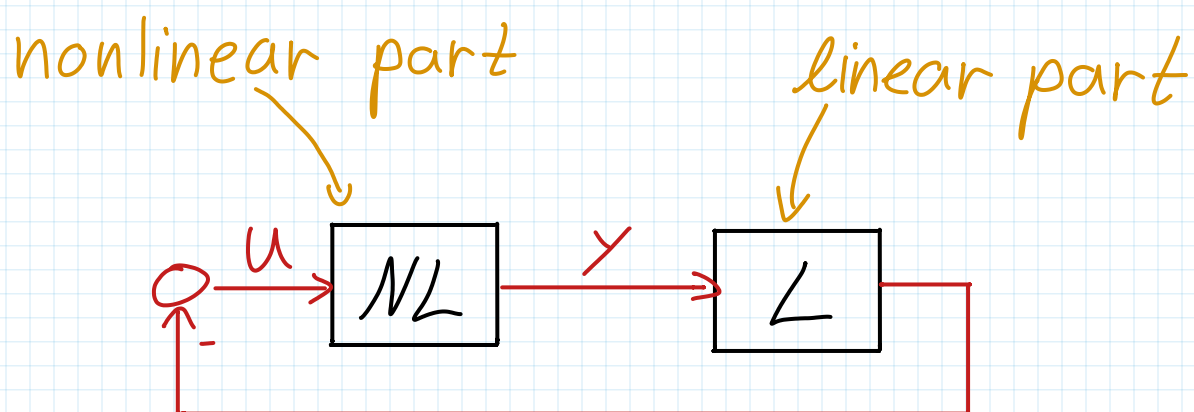


$\Rightarrow$ The error can grow arbitrarily large, but the input u can't!

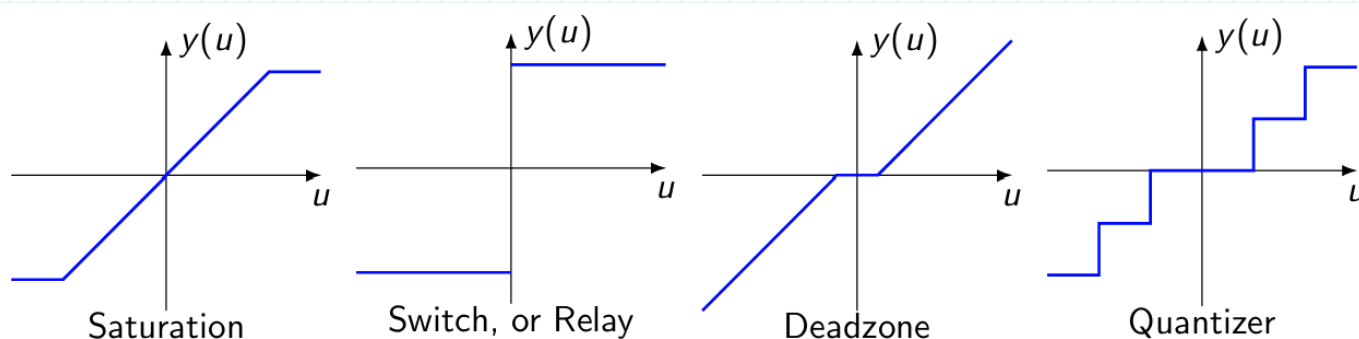
• Hence Sysiphus' strength limitations must be modeled as such:



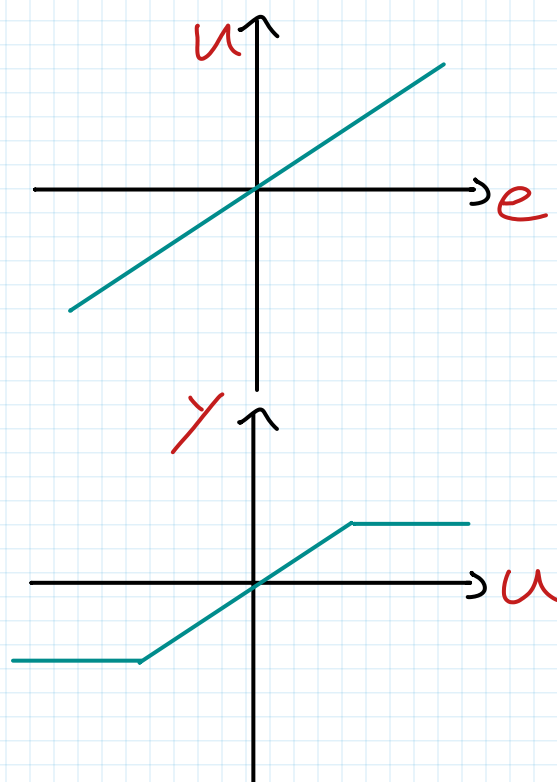
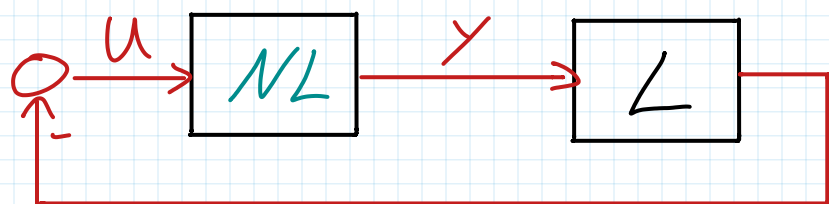
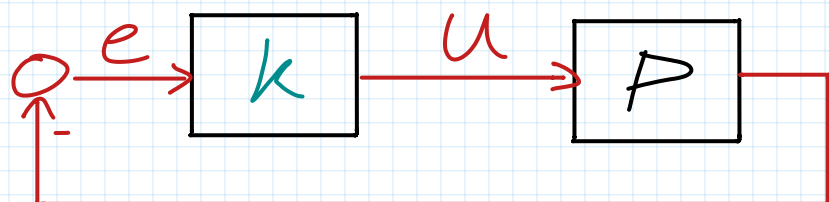
The trick will now be to model this nonlinear element separately and keep the linearized plant.



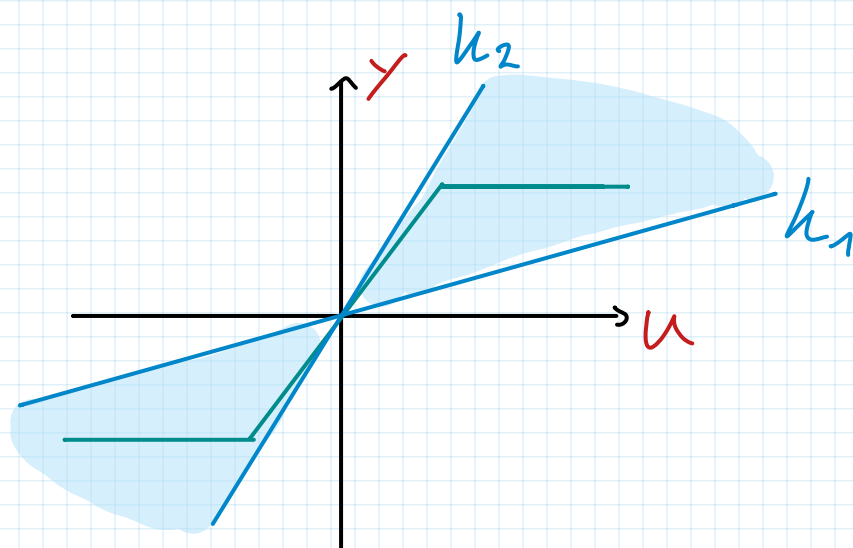
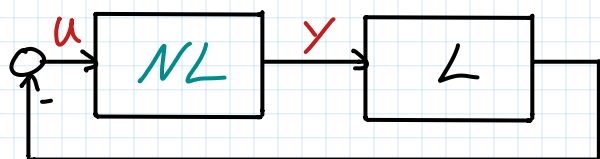
$\Rightarrow$ Common NL blocks include:



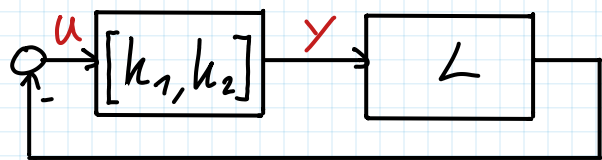
• The NL block acts like a nonlinear gain.



- Using this similarity to gains, we can assess stability of the system with the nonlinearity.
- A feedback system with a nonlinear element is absolutely stable if the closed loop is stable for all k that contain the possible nonlinearity sector.



- So now, if the loop is stable for any $k \in [k_1, k_2]$,



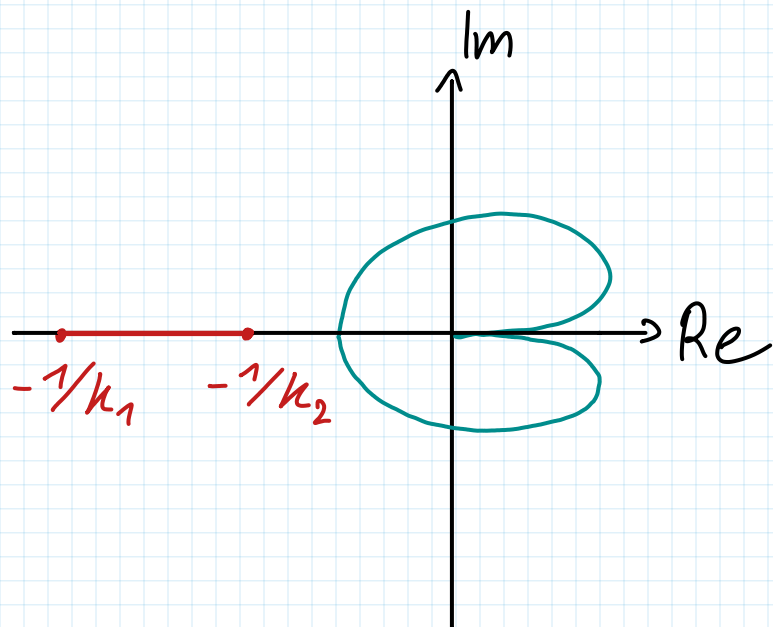
$\Rightarrow$ Its also stable with NL !

- Mathematically we need stability for:

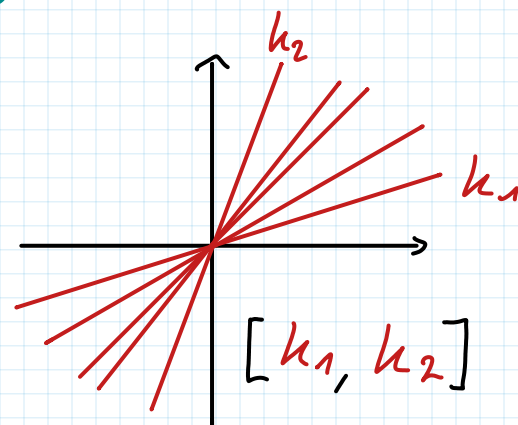
$$k_1 \cdot u \leq NL(u) \leq k_2 \cdot u$$

$\Rightarrow$ What does this condition look like on a Nyquist-plot?

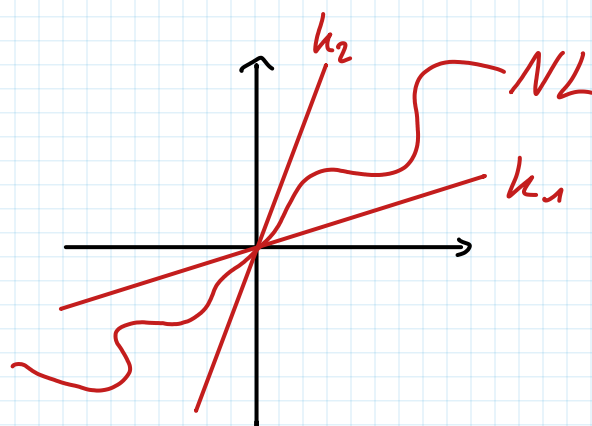
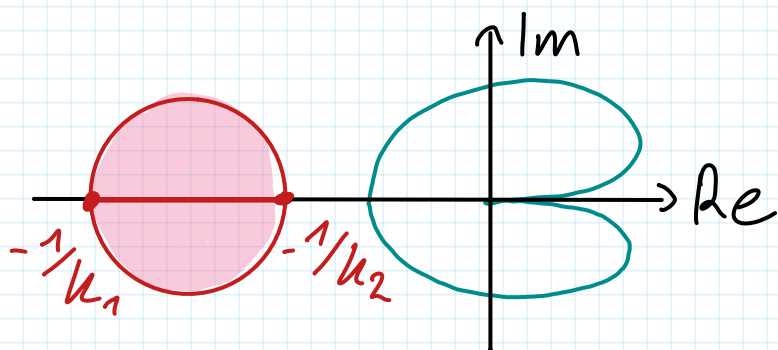
- We have an open loop stable $P(s)$ and $k_1 \leq k \leq k_2$



We now need to avoid encircling a segment of k 's!



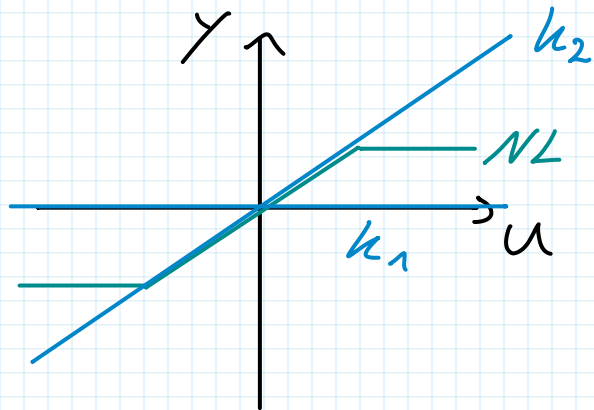
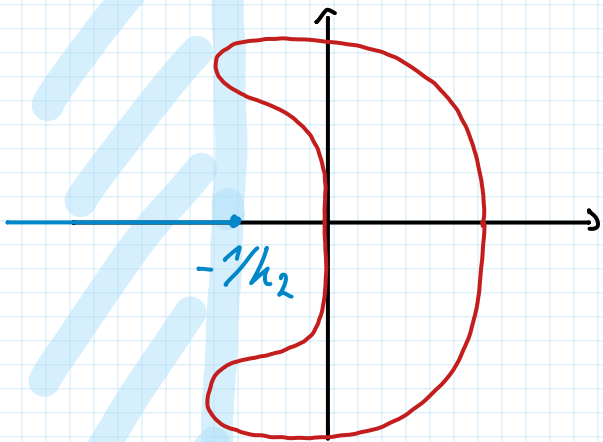
- For closed loop stability of the system with the nonlinearity, this is a necessary condition.
- The so called circle criterion is a sufficient condition:



$\Rightarrow$ For guaranteed stability, we must count encirclements of the circle.

Example:

• Assume an open-loop stable system with a saturation nonlinearity. The Nyquist plot is:



$\Rightarrow$ The Nyquist plot touches the circle!

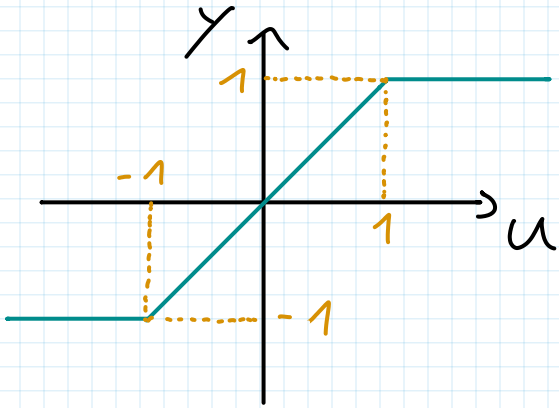
- Necessary condition fulfilled
- Sufficient condition violated

$\Rightarrow$ We can't conclude if the closed-loop is stable!

We need a more precise way to describe the nonlinearity...

Describing functions

- A saturation nonlinearity is written as:



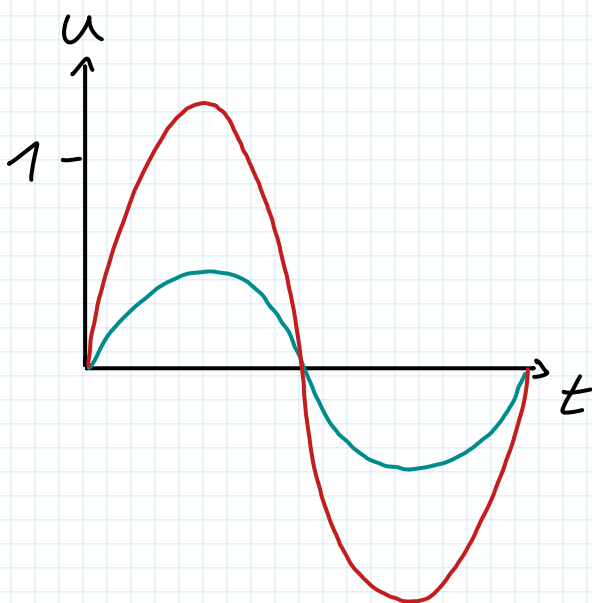
$$y = \begin{cases} 1, & u \geq 1 \\ u, & -1 < u < 1 \\ -1, & u \leq -1 \end{cases}$$

- In general our inputs u are sinusoids:

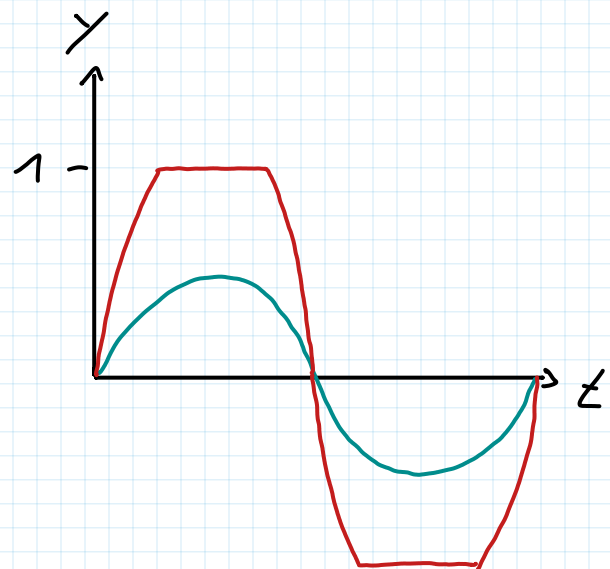
$$u(t) = A \sin(\omega t)$$

↖ Amplitude

- The effect of the saturation is dependent on the input amplitude A .



$\Rightarrow$
Sat

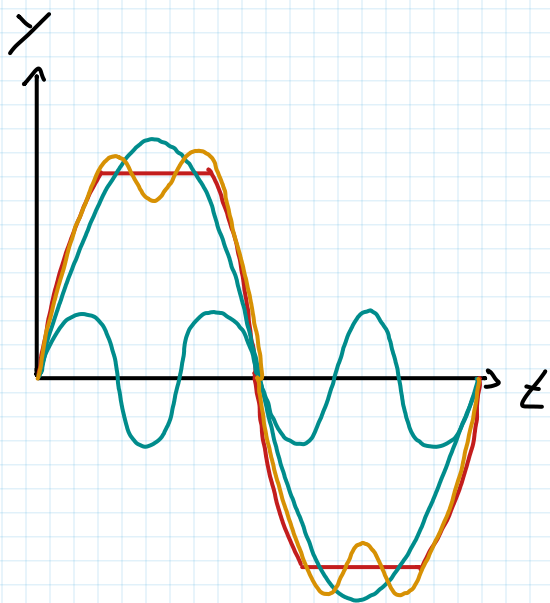


So the nonlinearity function

$$y(t) = \text{sat}(\underbrace{A \sin(\omega t)}_{u(t)}),$$

is also periodic if $u(t)$ is periodic.

What could we use to describe a function explicitly that has such a weird shape??



Periodic functions like this can be constructed by infinite sums of sinusoids!

$\Rightarrow$ Fourier series

We find the Fourier representation with:

$$y(t) = \frac{a_0}{2} + \sum_{i=1}^{\infty} (a_i \cos(i\omega t) + b_i \sin(i\omega t))$$

$$a_i = \frac{1}{\pi} \int_{-\pi}^{\pi} y(t) \cos(i\omega t) dt$$

$$b_i = \frac{1}{\pi} \int_{-\pi}^{\pi} y(t) \sin(i\omega t) dt$$

- For the saturation case, the NL is odd so all terms $a_i = 0$.
- In addition, we will approximate the NL-function with just the first coefficient b_1 . So the NL is described as:

$$y(t) \approx b_1 \sin(\omega t)$$

=> This simplification works, because real systems almost always reject high-frequency signals anyway.

So we find the coefficient b_1 as:

$$b_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} y(t) \sin(\omega t) dt,$$

input $u(t)$

where $y(t) = \text{sat}(A \sin(\omega t))$.

- clearly b_1 is dependent on amplitude A .

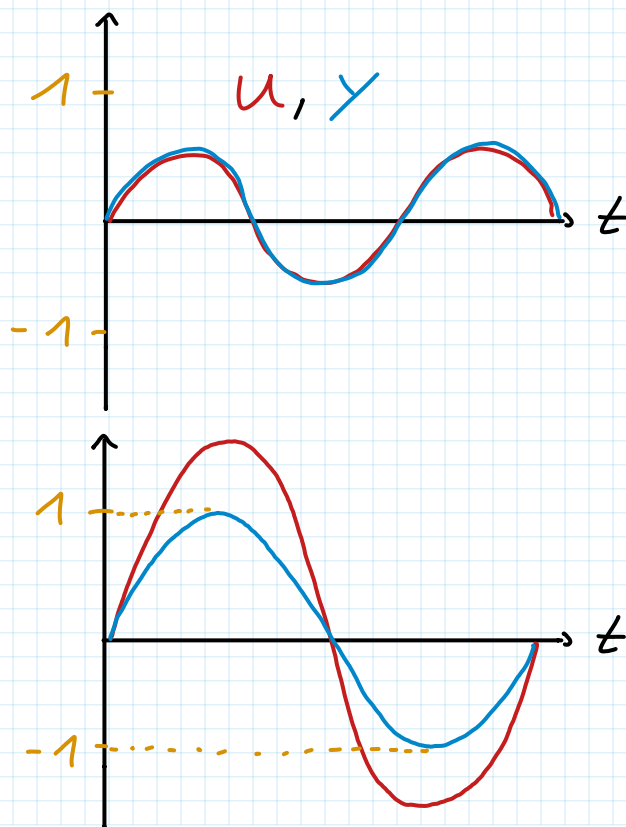
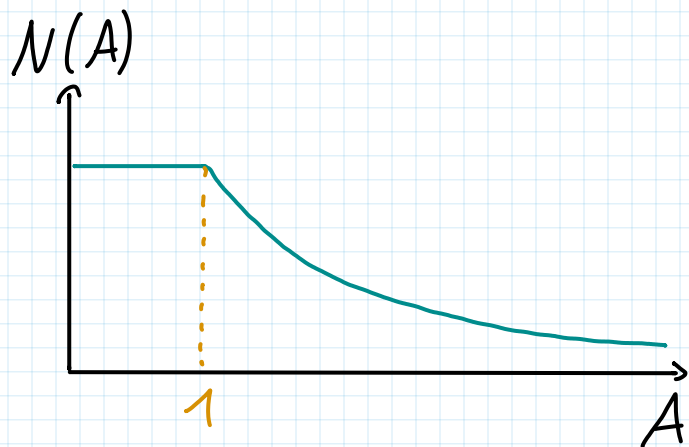
$$b_1(A) = \frac{1}{\pi} \int_{-\pi}^{\pi} \underbrace{\text{sat}(A \sin(\omega t))}_{y(t)} \underbrace{\sin(\omega t)}_{u(t)} dt$$

If we "normalize" $b_1(A)$ with A , we get the describing function of the saturation

$$N(A) = \frac{b_1(A)}{A} = \frac{1}{\pi A} \int_{-\pi}^{\pi} \gamma(t) \sin(\omega t) dt$$

If we solve this integral, we get different solutions depending on whether saturation occurs or not.

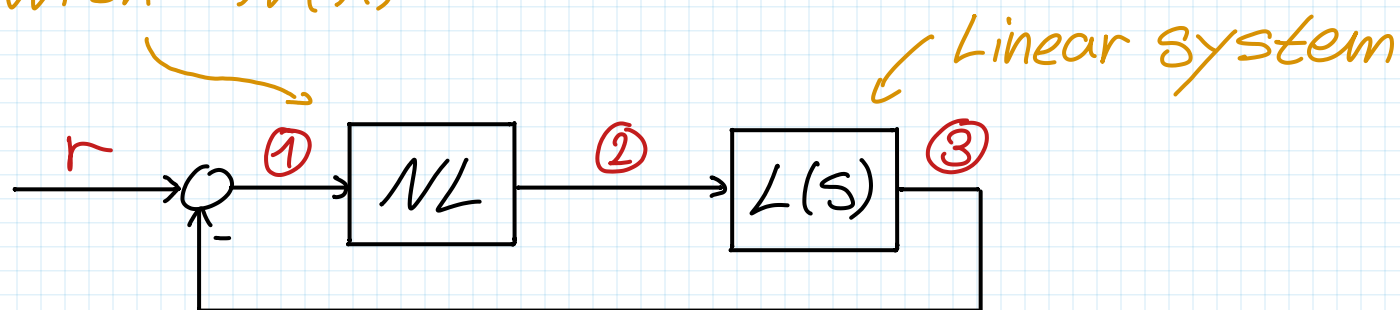
$$N(A) = \begin{cases} \frac{2}{\pi} \left(\sin^{-1}\left(\frac{1}{A}\right) + \frac{1}{A} \sqrt{1 - \left(\frac{1}{A}\right)^2} \right), & A > 1 \\ 1, & A \leq 1 \end{cases}$$



$\Rightarrow$ Here, $N(A)$ is a gain that decreases with amplitude.

Now we have a new method to describe nonlinearities. Can we now analyze stability using the describing functions?

Nonlinear system described with: $N(A)$



• We can now approximate the open-loop as

$$L'(A, s) \approx N(A)L(s)$$

Let's analyze how a signal propagates through this loop.

$$\Rightarrow \textcircled{1} = Ae^{st} \text{ (ass.)}$$

$$\Rightarrow \textcircled{2} = AN(A)e^{st}$$

$$\Rightarrow \textcircled{3} = AN(A)L(s)e^{st}$$

If we now set $r = 0$, does A

• Increase?

• Decrease?

• Not change?

Since we assumed that ① has constant amplitude, our system has a self-sustaining oscillatory behaviour if

$$\cancel{A}e^{\cancel{st}} = -\cancel{A}N(A)L(s)e^{\cancel{st}}$$

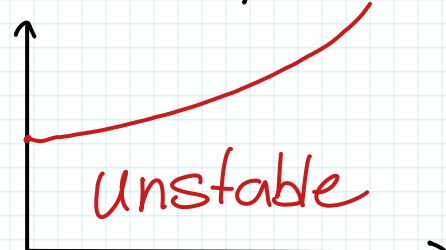
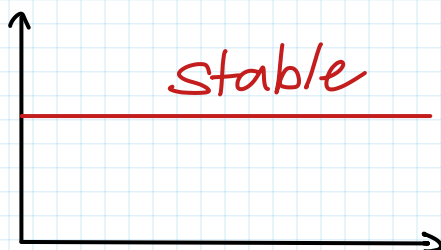
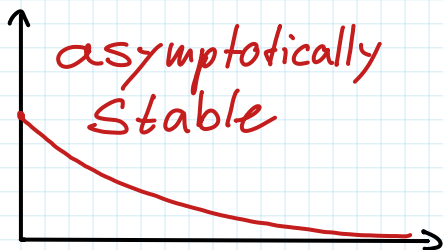
solving for the linear $L(s)$ we get:

$$L(s) = -\frac{1}{N(A)}$$

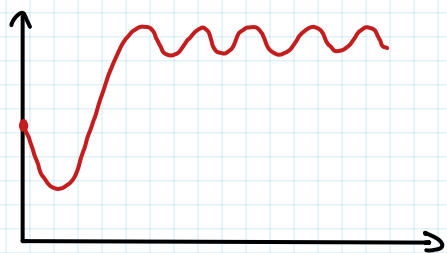
=> If this equation is fulfilled, a limit cycle is present.

This was very technical. Let's try and get an intuition for what a limit cycle is.

With NO reference inputs, our linear systems had 3 possible responses:

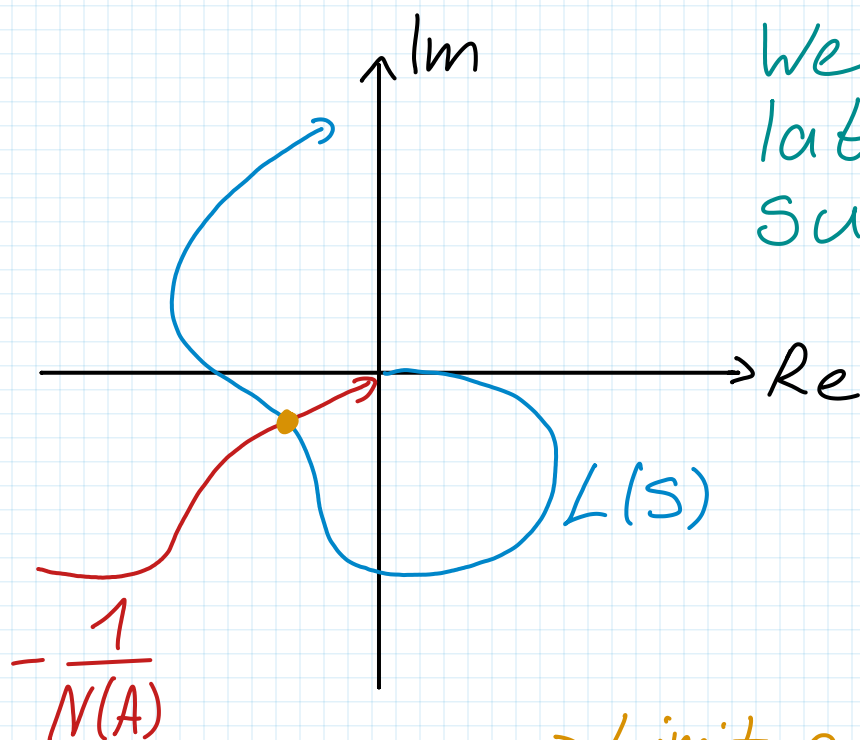


With nonlinear systems we can now also have stable, self sustaining oscillations:



Remember, there is no reference input! The system is doing this on its own!

• So how do we check for limit cycles?
 $\Rightarrow$ Check polar-plot / Nyquist:



We said the oscillation is self-sustaining if:

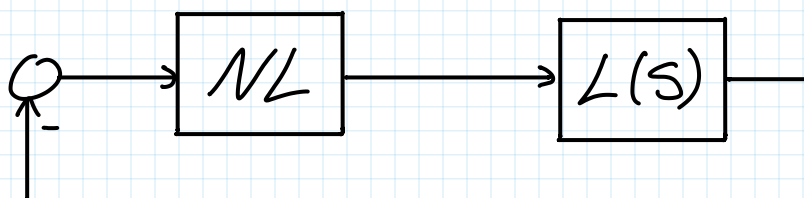
$$-\frac{1}{N(A)} = L(s)$$

$\Rightarrow$ Limit cycle at the intersection

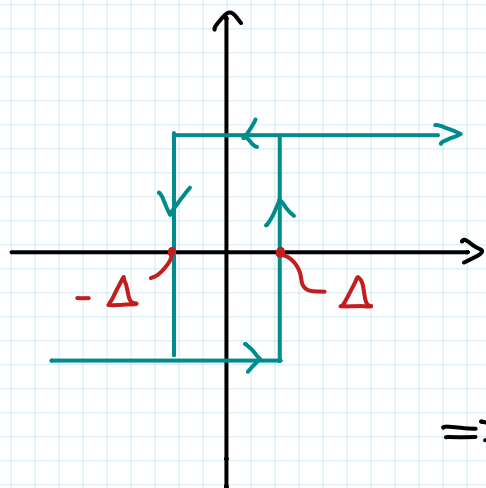
Example:

Schmitt-trigger

$$L(s) = \frac{1}{s(s+1)}$$



• Schmitt-trigger:

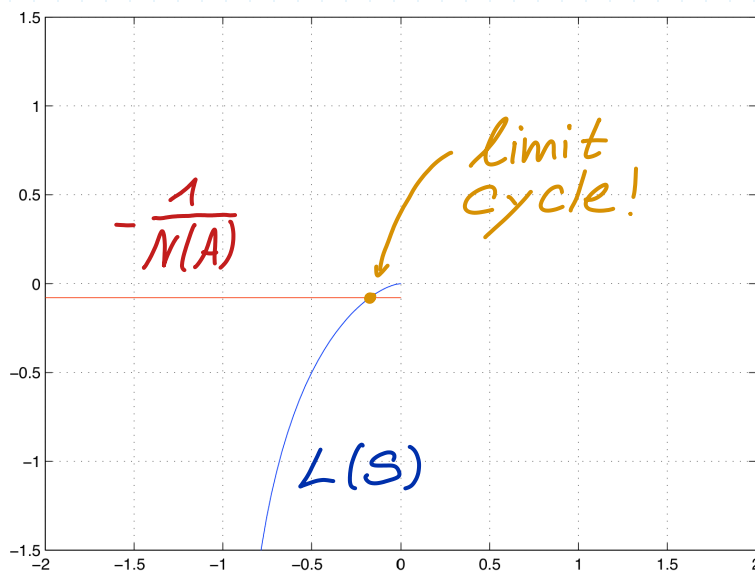


$$N(A) = \frac{4}{\pi A} \left(\sqrt{1 - \left(\frac{\Delta}{A} \right)^2} - j \frac{\Delta}{A} \right)$$

Δ : trigger value

$$\Rightarrow -\frac{1}{N(A)} = \frac{\pi}{4} \left(\sqrt{A^2 - \Delta^2} + j\Delta \right)$$

Drawing the polar plot:



• To find the ω and A that characterize the limit cycle we solve:

$$L(s) = -\frac{1}{N(A)} = \frac{\pi}{4} (\sqrt{A^2 - \Delta^2} + j\Delta)$$

$$\cdot L(j\omega) = \frac{1}{j\omega(j\omega+1)} = \underbrace{-\frac{1}{\omega^2+1}}_{\text{Real}} - j \cdot \underbrace{\frac{1}{\omega^3+\omega}}_{\text{Imag}}$$

Setting real and imaginary parts equal:

$$\cdot \omega^3 + \omega = \frac{4}{\pi \Delta}$$

Δ is given as:
 $\Delta = 0,1$

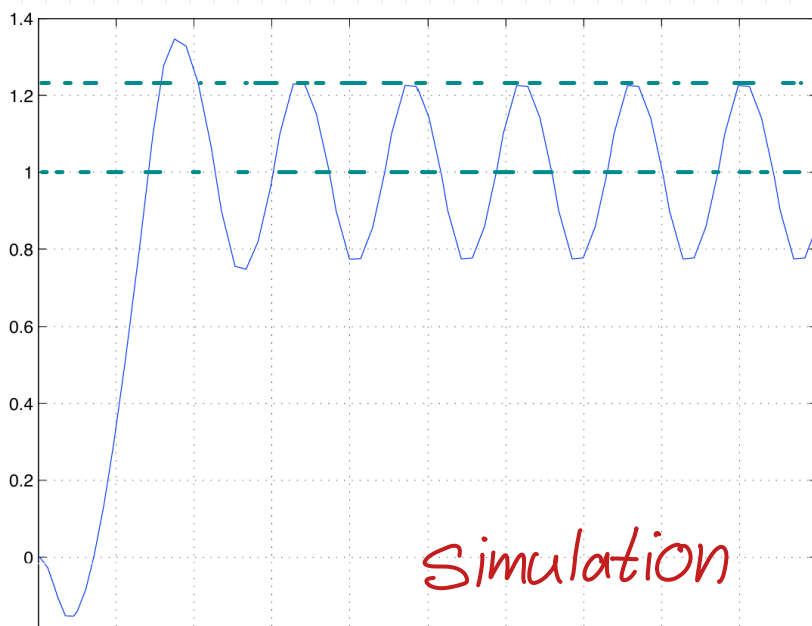
$$\cdot A^2 = \Delta^2 + \left(\frac{4}{\pi} \frac{1}{\omega^2+1} \right)^2$$

We get:

$$\omega = 2,2 \frac{\text{rad}}{\text{s}}$$

$$A = 0,24$$

A

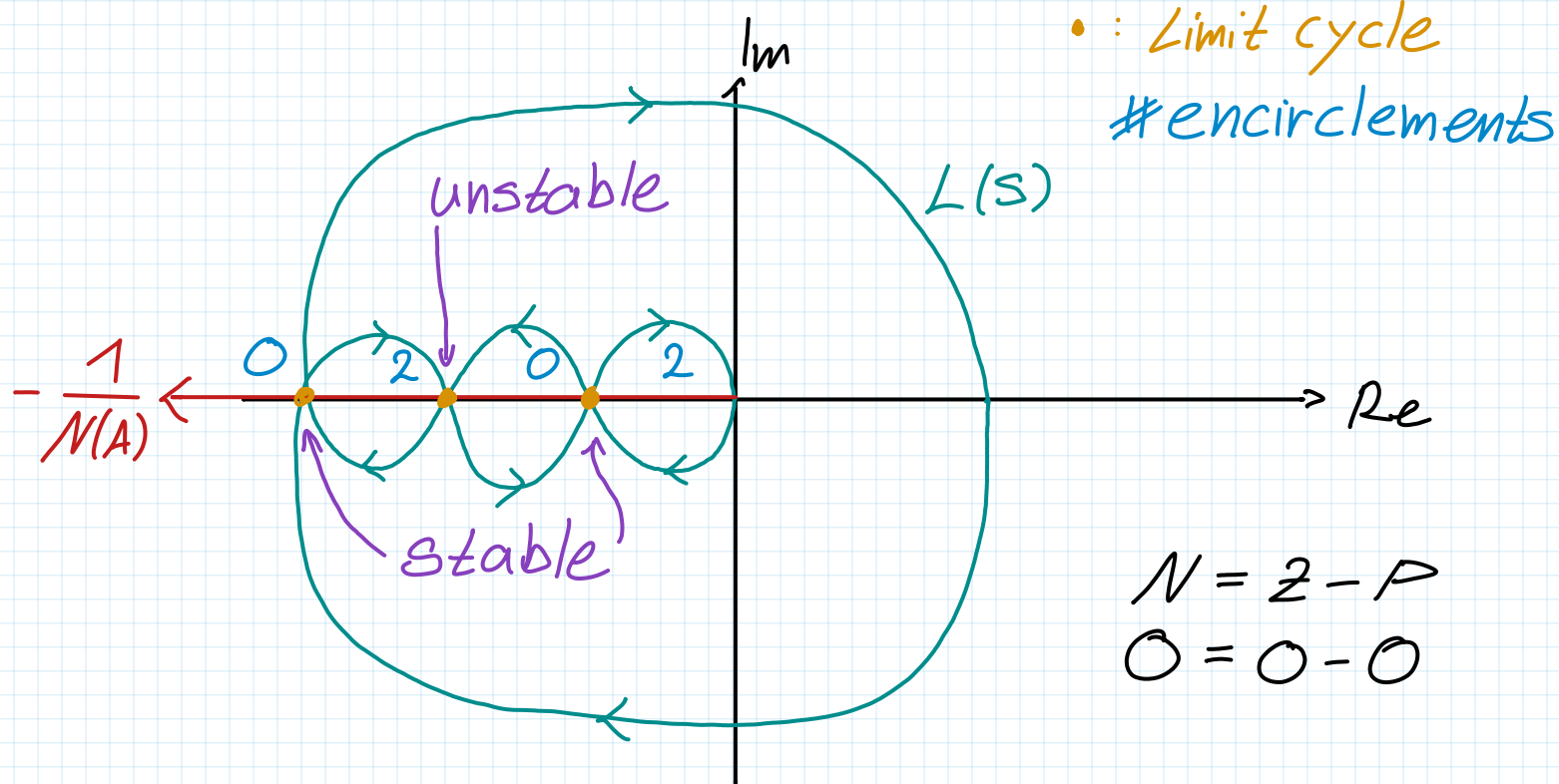


• More general: Limit cycles are stable if a small amplitude decrease causes a amplitude increase and vice-versa

=> We can also use the Nyquist plot to see if this is the case!

Example:

Given an open-loop stable system with Nyquist plot:



Let's look at an old exam question

Problem: Consider the closed-loop system T shown in Figure 23, where L is a linear time-invariant system and where NL is a saturation non-linearity.

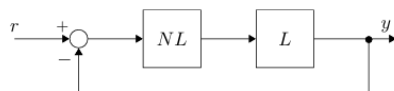


Figure 23: Standard feedback architecture. System L and saturation non-linearity denoted by NL .

The describing function $N(A)$ of the saturation non-linearity is given by,

$$N(A) = \frac{2}{\pi} \left[\arcsin\left(\frac{1}{A}\right) + \frac{1}{A} \sqrt{1 - \left(\frac{1}{A}\right)^2} \right].$$

It is known that $L(s)$ has a pole at the origin, i.e. at $s = 0$, and that $L(s)$ has one unstable pole. Figure 24 shows the Nyquist plot of $L(s)$ (solid line) together with $\frac{-1}{N(A)}$ (dashed line). Further, assume that all assumptions required for a describing function analysis are met.

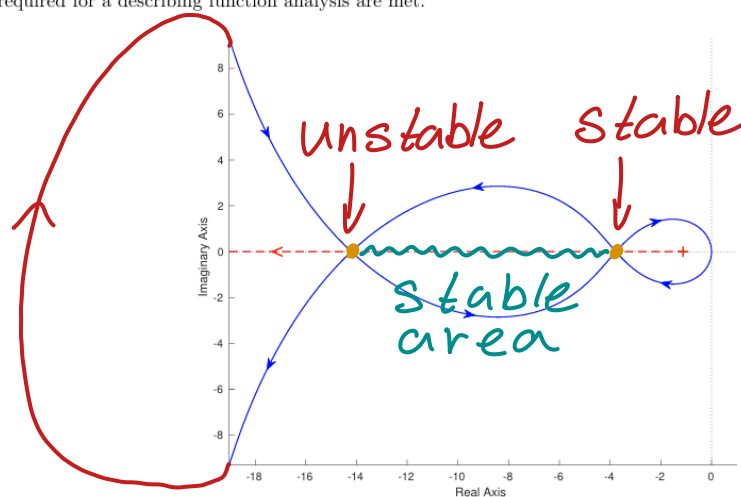


Figure 24: [Solid line] Nyquist plot of $L(s)$. [Dashed Line] Plot of $\frac{-1}{N(A)}$, where the arrow indicates the direction of $\frac{-1}{N(A)}$ as A increases.

Q40 (0.5 Points) Mark the correct answer.

First, without doing any stability analysis, what is the maximum number M of potential limit cycles that the closed-loop system T could support?

- ☐ A $M = 0$
☒ B $M = 2$

- ☐ C $M \rightarrow \infty$
☐ D $M = 1$

Q41 (1 Points) Mark the correct answer.

What is the number M_s of stable limit cycles that exist in the closed-loop system T ?

- ☐ A $M_s = 0$
☐ B $M_s = 2$

- ☐ C $M_s \rightarrow \infty$
☒ D $M_s = 1$

Q42 (1 Points) Mark all correct statements.

- ☐ A The circle criterion is a ~~necessary~~ and sufficient condition for absolute stability.
☒ B Describing functions can be seen as approximate transfer functions that represent the transfer function from sinusoidal inputs of amplitude A and frequency ω to the first harmonic of the output of the corresponding non-linearity.
☐ C Limit cycles are typically observed in linear systems.
☒ D The describing function of a static non-linearity is an amplitude dependent gain $N(A)$.

Finally, some quiz-questions!

$$N = \frac{0}{2} - 1$$

$$\Rightarrow N = -1$$

(1 CCW)

Q1) Describing functions also work well in systems with multiple nonlinearities

☐ TRUE ☒ FALSE

Q2) Describing functions are a good fit for systems that have high-pass-filter behavior.

☐ TRUE ☒ FALSE

Q3) Using describing functions makes the total open-loop function L' dependent on ω and A .

☒ TRUE ☐ FALSE

Have a nice weekend ü